

PH 322 – Lecture 23

Class Notes: Amplitude Amplification and Grover's Theorem

Dr. Shashank Gupta

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Reference Texts

- P. Kaye, R. Laflamme, and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press, Chapter 8.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

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1 The Search Problem and Quantum Speedup

We start from a very general task: searching for a “good” solution in a large, unstructured space.

1.1 Black-box search formulation

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a Boolean function marking good solutions:

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is a good solution,} \\ 0 & \text{otherwise.} \end{cases}$$

The **Search Problem** is:

Input: Oracle (black box) U_f that computes f .

Task: Find some $x \in \{0, 1\}^n$ such that $f(x) = 1$.

Classically, any algorithm that succeeds with constant probability on a black-box f must make $\Omega(N)$ queries, where $N = 2^n$. Grover’s quantum algorithm solves the same problem in $O(\sqrt{N})$ queries.

Grover’s Theorem (informal)

For unstructured search over $N = 2^n$ items with a unique solution, Grover’s algorithm succeeds with high probability using $O(\sqrt{N})$ oracle queries, which is optimal for quantum algorithms in the black-box model.

Toy Example:

Classical vs Quantum search Suppose $N = 1,000,000$. Classically, the expected number of queries to find the unique good solution is 500,000. Grover’s algorithm finds it in approximately $\frac{\pi}{4}\sqrt{10^6} \approx 785$ queries. That is a roughly $600\times$ reduction in the number of queries.

2 The Oracle and Phase Kickback

2.1 Standard oracle model

We assume an oracle of the form

$$U_f : |x\rangle |b\rangle \mapsto |x\rangle |b \oplus f(x)\rangle, \quad (1)$$

where $b \in \{0, 1\}$ and $\oplus$ denotes XOR. Setting the target qubit to $|0\rangle$, this simply computes $f(x)$.

2.2 Phase oracle via phase kickback

Prepare the target in

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Then

$$|x\rangle |-\rangle \xrightarrow{U_f} (-1)^{f(x)} |x\rangle |-\rangle.$$

Since the second register remains $|-\rangle$ (an eigenstate), we can ignore it, and view U_f as an n -qubit phase oracle:

$$U_f : |x\rangle \mapsto (-1)^{f(x)} |x\rangle. \quad (2)$$

Toy Example:

Phase kickback for a single-qubit oracle Let $n = 1$, $f(0) = 0$, $f(1) = 1$ (i.e. the unique good solution is $x = 1$). Apply U_f to the state $|+\rangle |-\rangle$:

$$|+\rangle |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |-\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2}}((-1)^0 |0\rangle + (-1)^1 |1\rangle) |-\rangle = |-\rangle |-\rangle.$$

The first qubit has flipped from $|+\rangle$ to $|-\rangle$, encoding f 's answer in a phase. This is phase kickback.

3 Grover's Algorithm: The 5-Element Toy Example

Throughout this section we work with a concrete search problem over $N = 8$ elements (nearest power of 2 enclosing 5; we will note where things differ for non-powers of 2) and then specialize to exactly track the $N = 8$ arithmetic.

Toy Example:

Setting up the 5-element search

We have an unstructured database of 5 elements labelled by their binary strings:

$$\{000, 001, 010, 011, 100\}.$$

We wish to find $w = 011$ (decimal 3), the unique good element. We pad to the nearest $N = 2^n$ to use the Hadamard trick; for simplicity of the amplitude calculations we work with $N = 8$ (3 qubits) and simply note that items $\{101, 110, 111\}$ are dummy elements that f marks as bad.

The oracle is:

$$f(x) = \begin{cases} 1 & x = 011, \\ 0 & \text{otherwise.} \end{cases}$$

Step 1: Initial state. Start with $|000\rangle$. Apply $H^{\otimes 3}$:

$$|\psi\rangle = H^{\otimes 3} |000\rangle = \frac{1}{\sqrt{8}} \sum_{x=0}^7 |x\rangle = \frac{1}{2\sqrt{2}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle).$$

Every element has amplitude $\alpha = 1/\sqrt{8} \approx 0.354$. The initial success probability is $1/8 = 12.5\%$.

3.1 Good and bad subspaces

Define:

$$|\psi_{\text{good}}\rangle = |w\rangle = |011\rangle, \quad |\psi_{\text{bad}}\rangle = \frac{1}{\sqrt{7}} \sum_{x \neq 011} |x\rangle.$$

The initial state decomposes as

$$|\psi\rangle = \frac{1}{\sqrt{8}} |011\rangle + \sqrt{\frac{7}{8}} |\psi_{\text{bad}}\rangle = \sin \theta |\psi_{\text{good}}\rangle + \cos \theta |\psi_{\text{bad}}\rangle,$$

where

$$\sin \theta = \frac{1}{\sqrt{8}} \approx 0.354, \quad \theta = \arcsin\left(\frac{1}{\sqrt{8}}\right) \approx 20.7^\circ.$$

Toy Example:

Decomposing the initial state for $N = 8$ Numerically:

$$\sin \theta = \frac{1}{\sqrt{8}} \approx 0.354, \quad p_{\text{good}} = \sin^2 \theta = \frac{1}{8} = 0.125.$$

$$\cos \theta = \sqrt{\frac{7}{8}} \approx 0.935, \quad p_{\text{bad}} = \cos^2 \theta = \frac{7}{8} = 0.875.$$

We want to rotate this state toward $|\psi_{\text{good}}\rangle$ so that $p_{\text{good}} \rightarrow 1$.

3.2 Real amplitude picture (Figures 8.4–8.6)

In the Mosca text, amplitudes are drawn as vertical bars above/below a horizontal axis. Before any Grover iteration, every bar has height $1/\sqrt{8} \approx 0.354$ and the mean amplitude is also $1/\sqrt{8}$ (Figure 8.4). Replace the dummy images below with the actual figures from the text.

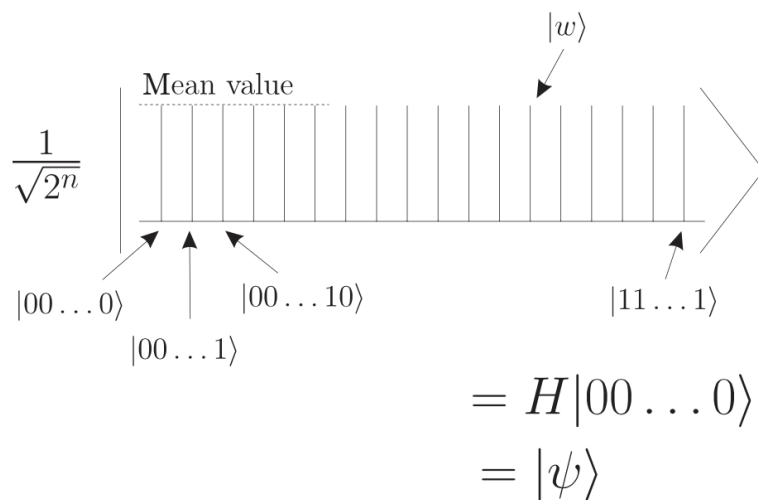


Figure 1: Uniform superposition: all 8 amplitudes equal to $1/\sqrt{8} \approx 0.354$. Replace with Fig. 8.4.

Toy Example:

Amplitude table before iteration

Basis state x	Amplitude α_x
000	+0.354
001	+0.354
010	+0.354
011 (w)	+0.354
100	+0.354
101	+0.354
110	+0.354
111	+0.354
Mean μ	+0.354

3.3 The Grover iterate (Figure 8.2)Define the n -qubit phase operator

$$U_0^\perp : \begin{cases} |0 \dots 0\rangle \mapsto |0 \dots 0\rangle, \\ |x\rangle \mapsto -|x\rangle, & x \neq 0 \dots 0, \end{cases}$$

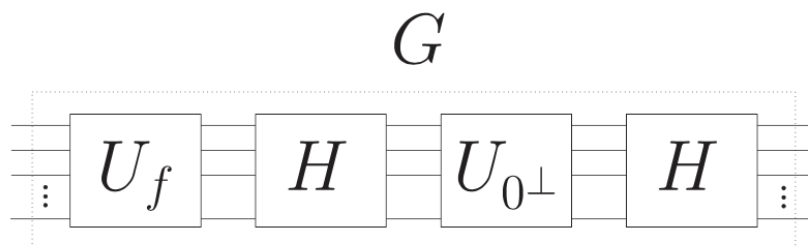
and let

$$U_\psi^\perp := H^{\otimes n} U_0^\perp H^{\otimes n} = 2|\psi\rangle\langle\psi| - I.$$

This is the **inversion about the mean**: every amplitude α_x is mapped to $2\mu - \alpha_x$ where μ is the arithmetic mean of all amplitudes.

The **Grover iterate** is

$$G = U_\psi^\perp U_f.$$

Figure 2: Circuit for the Grover iterate $G = U_\psi^\perp U_f$. Replace with Fig. 8.2.**3.4 First Grover iteration — step by step****Sub-step 1: Apply U_f (phase oracle)**

The oracle flips the sign of the good element:

$$U_f |\psi\rangle : \begin{cases} \alpha_{011} \rightarrow -\alpha_{011} = -0.354, \\ \alpha_x \rightarrow +\alpha_x = +0.354 & \forall x \neq 011. \end{cases}$$

The new mean is:

$$\mu' = \frac{1}{8} [7 \times (+0.354) + 1 \times (-0.354)] = \frac{6 \times 0.354}{8} = \frac{6}{8\sqrt{8}} \approx +0.265.$$

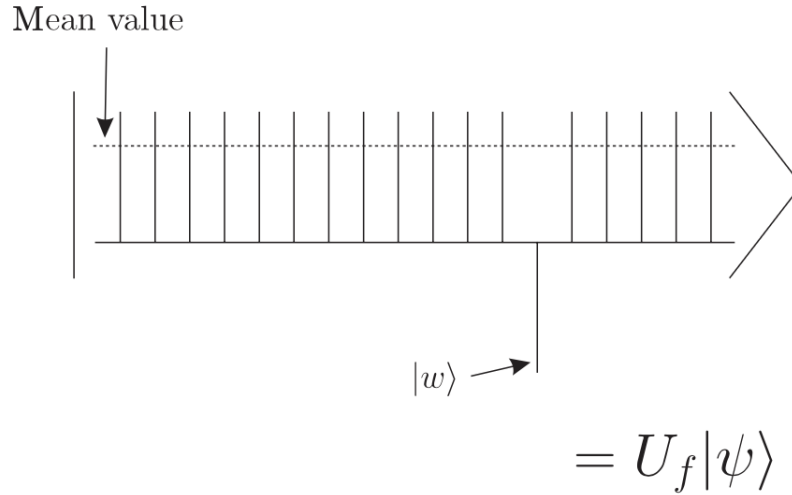


Figure 3: After U_f : amplitude of $|011\rangle$ is negated. Mean drops to ≈ 0.265 . Replace with Fig. 8.5.

Toy Example:

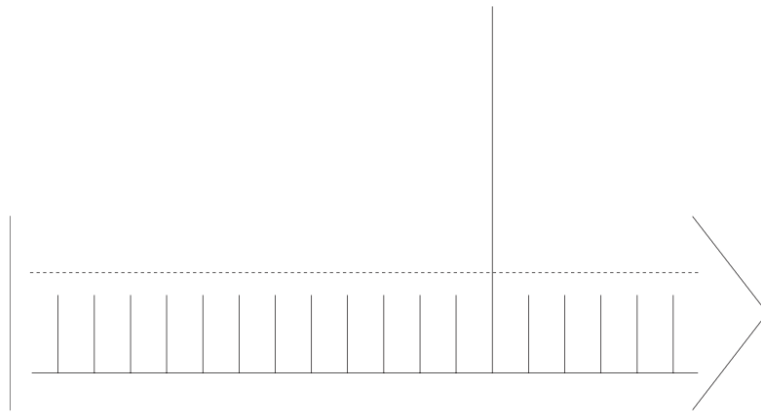
Amplitude table after U_f

x	α_x before	α_x after U_f
000	+0.354	+0.354
001	+0.354	+0.354
010	+0.354	+0.354
011	+0.354	-0.354
100	+0.354	+0.354
101	+0.354	+0.354
110	+0.354	+0.354
111	+0.354	+0.354
μ	+0.354	+0.265

Sub-step 2: Apply $U_\psi^\perp$ (inversion about the mean)

For each amplitude, apply $\alpha_x \mapsto 2\mu' - \alpha_x$:

- For $x \neq 011$: $2 \times 0.265 - 0.354 = 0.530 - 0.354 = +0.176$.
- For $x = 011$: $2 \times 0.265 - (-0.354) = 0.530 + 0.354 = +0.884$.



$$= U_{\psi^\perp} U_f |\psi\rangle$$

Figure 4: After inversion about the mean: amplitude of $|011\rangle$ jumps to ≈ 0.884 . Replace with Fig. 8.6.

Toy Example:

Amplitude table after one full Grover iterate G

x	After U_f	After $U_\psi^\perp U_f$
000	+0.354	+0.176
001	+0.354	+0.176
010	+0.354	+0.176
011	-0.354	+0.884
100	+0.354	+0.176
101	+0.354	+0.176
110	+0.354	+0.176
111	+0.354	+0.176

Success probability after 1 iteration:

$$P(\text{measure } 011) = (0.884)^2 \approx \mathbf{0.781}.$$

Starting from 12.5%, a single Grover iterate boosts the success probability to about 78.1%.

3.5 How many iterations?

Using the formula $k \approx \frac{\pi}{4}\sqrt{N}$ with $N = 8$:

$$k \approx \frac{\pi}{4}\sqrt{8} \approx \frac{3.14159}{4} \times 2.828 \approx 2.22.$$

So we apply $k = 2$ Grover iterations.

Toy Example:

Second Grover iteration for $N = 8$ After iteration 1, the amplitudes are: $\alpha_{011} = +0.884$, all others = $+0.176$.

Apply U_f :

$$\alpha_{011} \rightarrow -0.884, \quad \alpha_x \rightarrow +0.176 \quad (x \neq 011).$$

New mean:

$$\mu'' = \frac{1}{8} [7 \times 0.176 + (-0.884)] = \frac{1.232 - 0.884}{8} = \frac{0.348}{8} \approx 0.043.$$

Apply $U_\psi^\perp$ (inversion about μ''):

- For $x \neq 011$: $2 \times 0.043 - 0.176 = 0.086 - 0.176 = -0.090$.
- For $x = 011$: $2 \times 0.043 - (-0.884) = 0.086 + 0.884 = +0.970$.

Success probability after 2 iterations:

$$P(\text{measure } 011) = (0.970)^2 \approx \mathbf{0.941}.$$

After 2 iterations the probability of finding $w = 011$ is about 94%. This is close to the theoretical prediction from the formula $\sin^2((2k+1)\theta)$ with $k = 2$, $\theta \approx 20.7^\circ$:

$$\sin^2(5 \times 20.7^\circ) = \sin^2(103.5^\circ) \approx (0.972)^2 \approx 0.944.$$

4 The Geometric Picture: Two Reflections = One Rotation

4.1 The 2D subspace and angle θ

All of Grover's dynamics happens in the 2D subspace spanned by $\{|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle\}$. Define $\theta \in (0, \pi/2)$ by

$$\sin \theta = \frac{1}{\sqrt{N}}.$$

The orthogonal companion state to $|\psi\rangle$ inside this subspace is

$$|\psi^\perp\rangle = \cos \theta |\psi_{\text{good}}\rangle - \sin \theta |\psi_{\text{bad}}\rangle,$$

with zero mean amplitude.

After k Grover iterations, the state is:

$$G^k |\psi\rangle = \sin((2k+1)\theta) |\psi_{\text{good}}\rangle + \cos((2k+1)\theta) |\psi_{\text{bad}}\rangle. \quad (3)$$

Toy Example:

Verifying the rotation formula for $N = 8$ With $\theta = \arcsin(1/\sqrt{8}) \approx 20.7^\circ$:

k	$(2k+1)\theta$	$\sin^2((2k+1)\theta)$
0	20.7°	0.125 (initial)
1	62.1°	0.781
2	103.5°	0.944
3	144.9°	0.332 (starts to decrease)

The success probability oscillates sinusoidally. The optimal choice is $k = 2$ for $N = 8$. Beyond that, the state overshoots $|\psi_{\text{good}}\rangle$ and starts rotating back.

4.2 Geometric view: two reflections

The Grover iterate $G = U_\psi^\perp U_f$ is the composition of two reflections in the 2D subspace:

- U_f : reflection about the axis $|\psi_{\text{bad}}\rangle$ (flips the sign of the component along $|\psi_{\text{good}}\rangle$).
- $U_\psi^\perp$: reflection about the axis $|\psi\rangle$ (the initial state).

Two reflections whose axes meet at angle θ produce a rotation by 2θ . This is why each Grover iterate rotates the state by exactly 2θ toward $|\psi_{\text{good}}\rangle$.

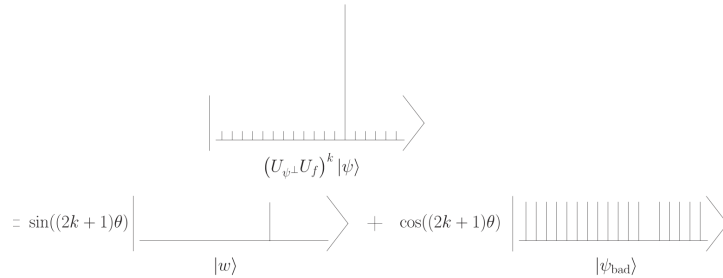


Figure 5: Geometric picture: G rotates the state by 2θ in the $|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle$ plane. Replace with the sequence Figs. 8.7–8.13 from the text.

Toy Example:

Rotation angle for $N = 8$

$$2\theta = 2 \arcsin\left(\frac{1}{\sqrt{8}}\right) \approx 41.4^\circ.$$

Starting at $\theta \approx 20.7^\circ$ from $|\psi_{\text{bad}}\rangle$, after 2 rotations of 41.4° we are at $20.7^\circ + 2 \times 41.4^\circ = 103.5^\circ$ from $|\psi_{\text{bad}}\rangle$, which is $103.5^\circ - 90^\circ = 13.5^\circ$ past $|\psi_{\text{good}}\rangle$ — explaining why the probability is 0.944 and not 1.

5 Grover's Search Algorithm

Grover's Quantum Search Algorithm

Let $N = 2^n$, and assume exactly one w satisfies $f(w) = 1$. Then:

1. Start with $|0\rangle^{\otimes n}$.
2. Apply $H^{\otimes n}$ to prepare

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

3. Apply the Grover iterate $G = U_{\psi}^{\perp} U_f$ a total of

$$k = \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor$$

times.

4. Measure in the computational basis. The outcome is w with probability $1 - O(1/N)$.

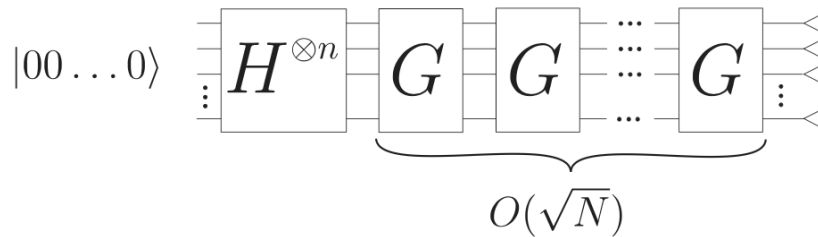


Figure 6: Schematic of Grover's search algorithm. Replace with Fig. 8.3.

Toy Example:

Complete run for $N = 8$, searching for $w = 011$

1. **Initialize:** $|000\rangle$.
2. **Superposition:** Apply $H^{\otimes 3}$. All 8 amplitudes $= 1/\sqrt{8} \approx 0.354$. Success probability $= 12.5\%$.
3. **Iterate $k = 2$ times:**
 - After iteration 1: amplitude of $|011\rangle = 0.884$, all others $= 0.176$. Success probability $= 78.1\%$.
 - After iteration 2: amplitude of $|011\rangle = 0.970$, all others $= -0.090$. Success probability $= 94.1\%$.
4. **Measure:** Output $|011\rangle$ with probability 94.1% .

Classically, searching through 8 items takes an expected 4.5 queries. Grover uses 2 — the quadratic speedup is already visible.

6 From Grover to Amplitude Amplification

Grover's algorithm is a special case of a more general procedure where the initial preparation algorithm A can be arbitrary.

6.1 General setting

Let A be any quantum algorithm that maps $|0\dots 0\rangle$ to

$$|\psi\rangle = A|0\dots 0\rangle = \sum_x \alpha_x |x\rangle |\text{junk}(x)\rangle, \quad (4)$$

with good success probability

$$p_{\text{good}} = \sum_{x \in X_{\text{good}}} |\alpha_x|^2 = \sin^2 \theta.$$

The generalized iterate is

$$Q = AU_0^\perp A^{-1}U_f,$$

which generalizes G by replacing $H^{\otimes n}$ with A .

6.2 Rotation formula for general A

Exactly as in Grover's special case:

$$Q^k |\psi\rangle = \sin((2k+1)\theta) |\psi_{\text{good}}\rangle + \cos((2k+1)\theta) |\psi_{\text{bad}}\rangle. \quad (5)$$

The optimal number of iterations is $k = \Theta(1/\sqrt{p_{\text{good}}})$.

Amplitude Amplification

If algorithm A guesses a good solution with probability $p_{\text{good}} > 0$, then $O(1/\sqrt{p_{\text{good}}})$ calls to the iterate Q and the oracle U_f produce a good solution with constant probability. This is a quadratic improvement over classical repetition, which requires $\Theta(1/p_{\text{good}})$ trials.

Toy Example:

Amplitude amplification with a biased initial algorithm Suppose A is a heuristic algorithm that guesses the solution with probability $p_{\text{good}} = 1/4$ (instead of uniform $1/N$).

- $\sin \theta = \sqrt{1/4} = 1/2 \implies \theta = 30^\circ$.
- Optimal iterations: $k \approx \pi/(4 \times 30^\circ \times \pi/180) = \pi/4 \times (180/30)/\pi = 1.5 \implies k = 1$.
- After 1 iteration: success probability $= \sin^2(3\theta) = \sin^2(90^\circ) = 1$.

With $p_{\text{good}} = 1/4$, a single application of Q makes the algorithm deterministic! This is much better than classical repetition, which would require 4 trials on average.

Toy Example:

Amplitude amplification with $p_{\text{good}} = 1/2$ Let $\sin^2 \theta = 1/2 \implies \theta = 45^\circ$.

Optimal iterations: $k \approx \pi/(4 \times 45^\circ \times \pi/180) = 1$.

After $k = 1$:

$$\sin^2(3 \times 45^\circ) = \sin^2(135^\circ) = (\sin 45^\circ)^2 = 1/2.$$

One iteration does not improve things here; the state rotates past $|\psi_{\text{good}}\rangle$. This is a corner case demonstrating that the formula $k = \lfloor \pi/(4\theta) \rfloor$ must be applied carefully.

6.3 Rotation picture (Figure 8.14–8.19)

In the $\{|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle\}$ plane, the state traces out an arc on the unit circle as Grover iterates are applied:

Explicitly:

- U_f is a reflection about the $|\psi_{\text{bad}}\rangle$ axis.
- $U_\psi^\perp$ is a reflection about the $|\psi\rangle$ axis.
- The product $Q = U_\psi^\perp U_f$ is therefore a rotation by 2θ toward $|\psi_{\text{good}}\rangle$.

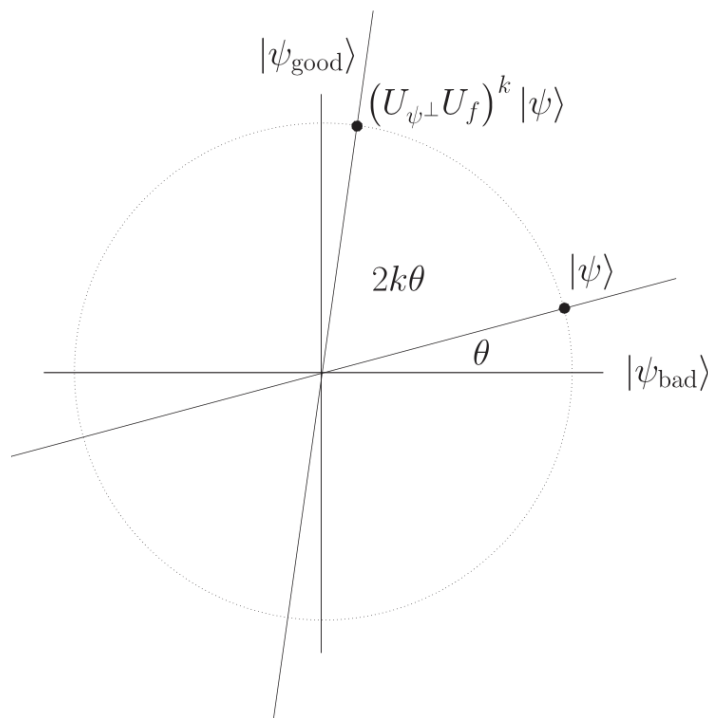


Figure 7: Amplitude amplification as rotation on the unit circle in the $|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle$ plane. Replace with Figs. 8.14–8.19 from the text.

7 Complexity and Optimality

7.1 Single-solution search

For $N = 2^n$ with one marked item:

$$p_{\text{good}} = \frac{1}{N}, \quad k = \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor, \quad P_{\text{success}} = 1 - O\left(\frac{1}{N}\right).$$

7.2 Multiple solutions

If there are t solutions, the probability in the initial state is $p_{\text{good}} = t/N$, and the optimal number of Grover iterations is $k = \Theta(\sqrt{N/t})$.

Toy Example:

Multiple solutions: 3 out of 8 Suppose 3 of 8 items are good. Then $p_{\text{good}} = 3/8$, $\sin^2 \theta = 3/8$, $\theta \approx 37.8^\circ$.

Optimal $k \approx \pi / (4 \times 37.8^\circ \times \pi / 180) \approx 1.2$, so $k = 1$.

After 1 iteration:

$$P_{\text{success}} = \sin^2(3\theta) = \sin^2(113.4^\circ) \approx (0.917)^2 \approx 0.841.$$

Classically, the expected number of queries is $N/t = 8/3 \approx 2.67$. Grover finds a solution with probability 84% in just 1 query.

7.3 General amplitude amplification

For general A with success probability p_{good} , amplitude amplification turns A into an algorithm using $O(1/\sqrt{p_{\text{good}}})$ applications vs. classical $\Theta(1/p_{\text{good}})$.

8 Conceptual Summary

- Grover's algorithm works by two reflections: (1) negate the good state (oracle), (2) invert about the mean. Their composition is a rotation in the 2D subspace toward $|\psi_{\text{good}}\rangle$.
- In the $\{|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle\}$ basis, each iterate rotates by angle 2θ , and after $k \approx \pi/(4\theta)$ steps the state is nearly $|\psi_{\text{good}}\rangle$.
- Amplitude amplification generalizes Grover search to any guessing algorithm A , providing a systematic quadratic speedup over classical repetition.
- The $N = 8$ toy example shows:
 - Start: 12.5% success probability.
 - After 1 iterate: 78.1%.
 - After 2 iterates: 94.1%.
 - Optimal classical search: 4.5 queries expected vs. 2 Grover iterates.

Exercises

Exercise 1 (Phase oracle equivalence). Derive the phase oracle Eq. (2) from the standard oracle in Eq. (1) by setting the target qubit to $|-\rangle$.

Exercise 2 (Reflection about the mean). Show that $U_{\psi}^{\perp} = H^{\otimes n} U_0^{\perp} H^{\otimes n} = 2|\psi\rangle\langle\psi| - I$, and prove that this maps each amplitude α_x to $2\mu - \alpha_x$ where μ is the mean.

Exercise 3 (Rotation angle). For $N = 8$ and target $w = 011$, compute θ explicitly and verify numerically that the amplitude of $|w\rangle$ after $k = 2$ Grover iterations matches $\sin(5\theta)$.

Exercise 4 (N=16 search). For $N = 16$ with 1 marked item:

- Compute the initial amplitude α_0 , angle θ , and success probability.
- Determine the optimal number of Grover iterations k .
- Fill in an amplitude table analogous to the $N = 8$ toy example, tracking iterations 1 through k .

Exercise 5 (Multiple solutions). For $N = 16$ with $t = 4$ marked items, compute θ , determine the optimal k , and find the success probability after k iterations.

Exercise 6 (General amplitude amplification). Let $A|0\dots 0\rangle = \sin\theta|\psi_{\text{good}}\rangle + \cos\theta|\psi_{\text{bad}}\rangle$. Show that $Q = AU_0^{\perp}A^{-1}U_f$ acts as a rotation by 2θ in the 2D subspace, leading to Eq. (5).

Exercise 7 (Quadratic speedup). Suppose a classical algorithm guesses a solution with probability $p > 0$ per trial.

- (a) Show the expected number of trials is $1/p$.
- (b) Compare to the $O(1/\sqrt{p})$ queries needed by amplitude amplification.
- (c) For $p = 1/100$, how many queries does each method need?

Exercise 8 (Exact solution with $p_{\text{good}} = 1/4$). Suppose A produces a good solution with probability $1/4$. Compute θ , determine $k = 1$, and verify that $\sin^2(3\theta) = 1$. Why does a single iterate give exact success in this case?

Exercise 9 (Overshooting). For $N = 4$ and 1 marked item, compute $\theta = 30^\circ$. Show that after $k = 1$ iteration the success probability is exactly 1, but after $k = 2$ it drops. What does this tell you about choosing the right number of Grover iterations?

Exercise 10 (NP-complete problems). Many NP-complete problems can be cast as unstructured search over a space of candidate solutions with a polynomial-time checker. Explain why Grover's algorithm gives a quadratic speedup in this context, but does not by itself imply an exponential speedup for NP-complete problems.

References

- [1] P. Kaye, R. Laflamme, and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press.
- [2] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.
- [3] L. K. Grover, "A Fast Quantum Mechanical Algorithm for Database Search," *Proceedings of the 28th ACM STOC*, 1996.